

Lecture 1

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1 Error

1.1 Approximation

1.2 Roundoff Error

Usually a machine uses a finite amount of memory to store a number.

machine $\leftarrow \pi$

Suppose a machine can store 10 digits.

1. Allocate a digit for sign, positive or negative.
2. Allocate a number of digits for the mantissa.
3. Allocate a number of digits for the exponent.
4. Allocate a digit for sign of the exponent.

Under rounding in the case of a 4 digit mantissa, $\pi = 0.3142 \cdot 10^1$, this is the machine # representation of pi.

Def: Roundoff error are all errors arising from the error between actual # and machine #.

Notation:

$$x = \text{real } \# fl(x) = \text{machine } \# \text{ representation of } x$$

In an s-digit rounding machine,

$$\text{Abs error} = |x - fl(x)|$$

$$\text{Rel error} = \frac{|x - fl(x)|}{|x|} \leq \frac{1}{2} \cdot 10^{1-5} = 5 \cdot 10^{-5}$$

1.3 Floating Point Arithmetic

What happens under arithmetic operations? (+, -, ·, /)

Two things to watch out for:

1. Many operations can degrade error.
2. Catastrophic cancellation.

number	machine
x	$fl(x)$
y	$fl(y)$
$x + y$	$fl(fl(x) + fl(y))$

The same follows for -, ·, /.

When performing arithmetic, it temporarily allows for more digits than the mantissa allows.

1.3.1 Catastrophic Cancellation

Ex: 3-digit rounding machine. Given 2 numbers: 0.312 and 0.311.

$$0.312 - 0.311 = 0.001 = 0.1 \cdot 10^{-2} \text{ Loss of significant digits!}$$

Ex: 5-digit rounding machine. Given 2 numbers:

Algorithmic "solutions"

$$((x + y) + z) + w$$

Worst case, 4 times the error.

$$(x + y) + (z + w)$$

Worst case, 3 times the error.